Supplementary Materials: Per-seed reliability diagrams for the 60-experiment transfer grid

Supplementary Materials: S1. Per-seed reliability diagrams

This section catalogs the 60 per-seed reliability diagrams referenced in the main text. Each figure shows the calibration curve, reliability diagram, and ΔECE bar plot for one of the 60 seed experiments (6 transfer pairs × 2 architectures × 5 seeds). Figures are named `fig_reliability_supp_{pair}_{arch}_seed{s}.pdf` and are included as external files with the submission.

S1. Per-seed reliability diagrams

The 60 figures are organized by transfer pair and architecture. Each figure corresponds to one row in the master results table and provides the visual calibration evidence underlying the ΔECE values reported in the main text. For each direction, ResNet1D and InceptionTime reliability diagrams are shown across seeds 42–46.

S1.1. Chapman→CPSC

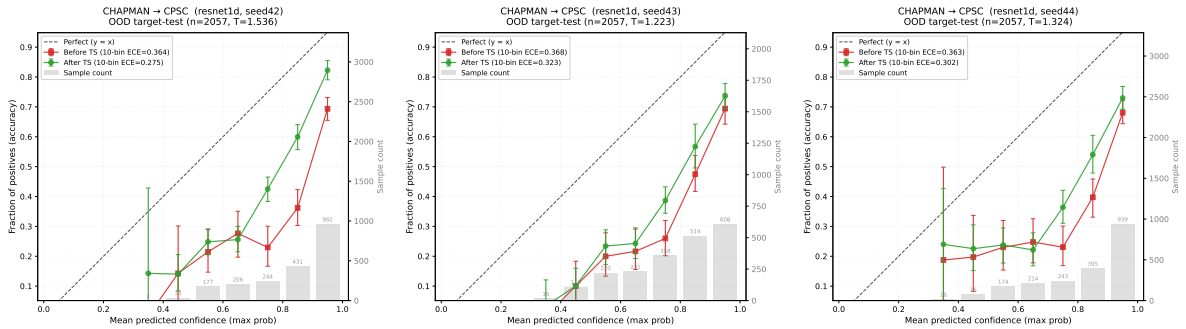


Figure 1: Chapman→CPSC reliability diagrams: ResNet1D seeds 42–44.

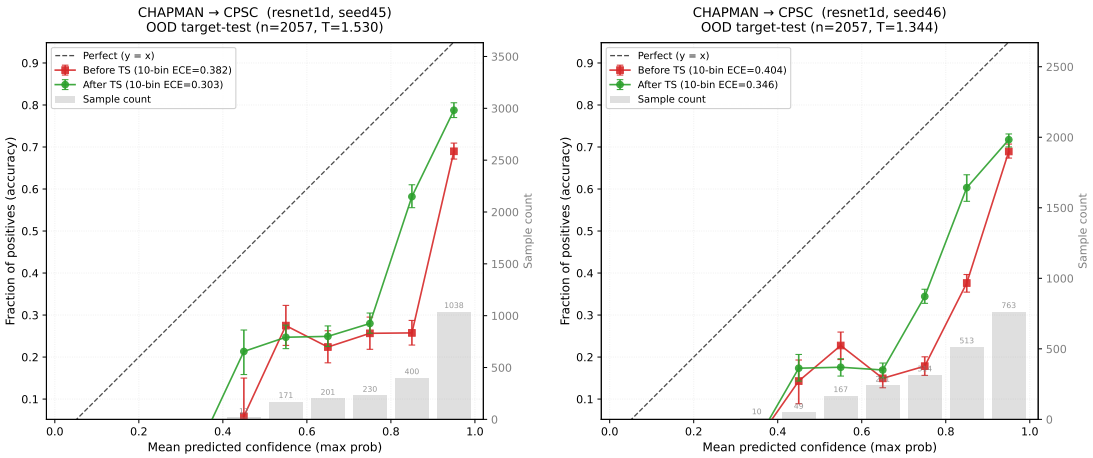


Figure 2: Chapman→CPSC reliability diagrams: ResNet1D seeds 45–46.

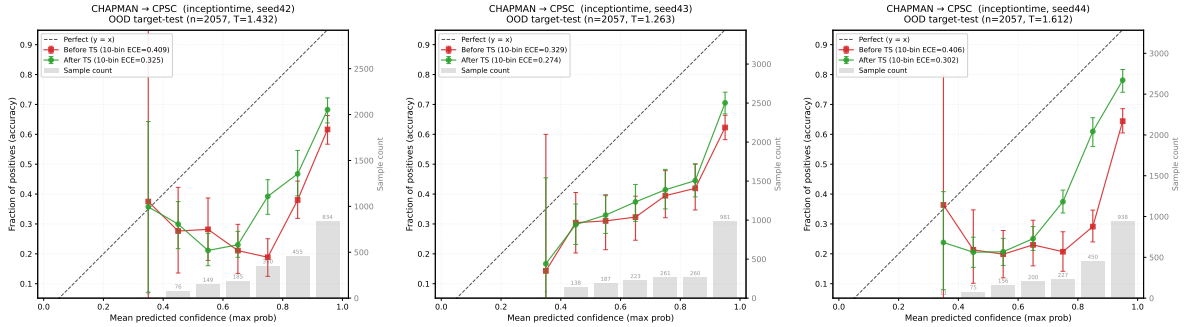


Figure 3: Chapman→CPSC reliability diagrams: InceptionTime seeds 42–44.

S1.2. Chapman→PTB-XL

S1.3. CPSC→Chapman

S1.4. CPSC→PTB-XL

S1.5. PTB-XL→Chapman

S1.6. PTB-XL→CPSC

S2. Additional supplementary materials

- **Per-experiment result files**: full JSON results (`checkpoints/transfer/**/transfer_result.json`) and CSV summaries (`results/*.csv`) for all 60 seed experiments.
- **Pre-registered protocol**: immutable public archive with per-file SHA-256 checksums, including the revision A1 full text (`https://doi.org/10.17605/OSF.IO/53H64`).
- **NCV experiment data**: E3 CSV containing `ncv_id` and `ncv_ood` columns for the net clinical value experiment (main text, decision-curve analysis subsection).
- **Shapley decomposition cells**: 27-cell validation table for the Shapley attribution analysis at $n = 20{,}000$.

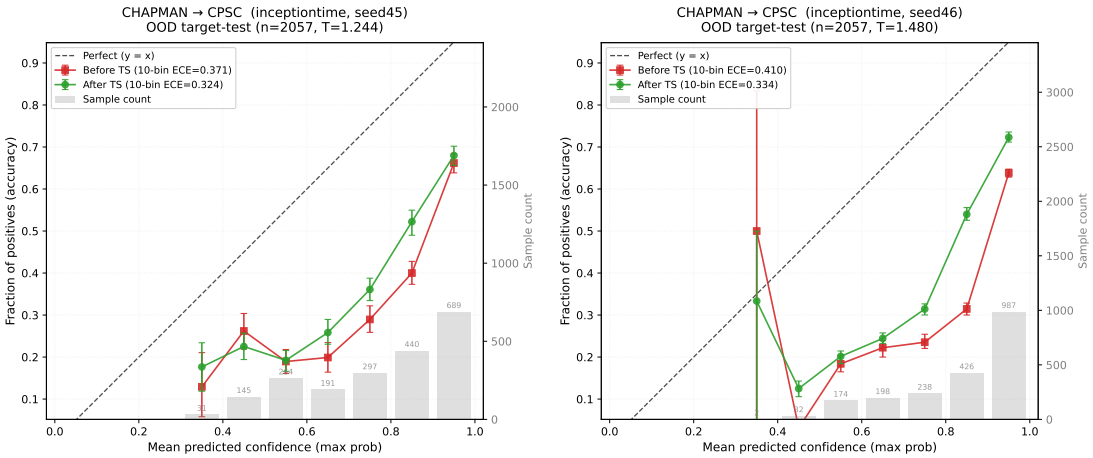


Figure 4: Chapman→CPSC reliability diagrams: InceptionTime seeds 45–46.

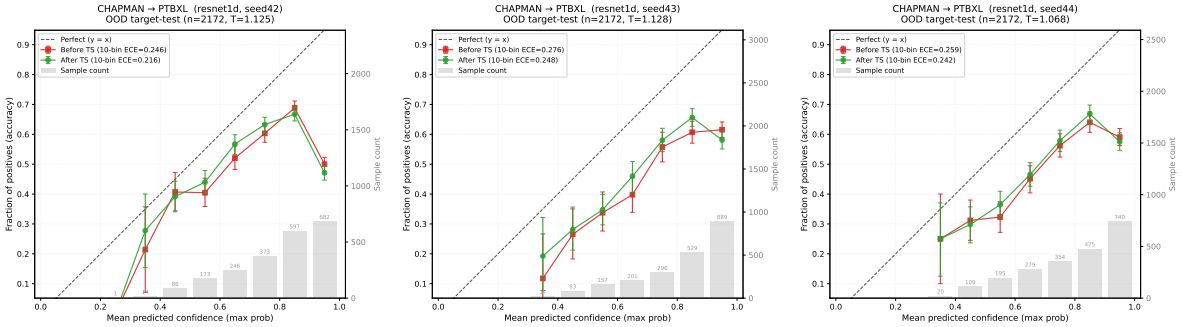


Figure 5: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

- **BCa CI rerun manifest**: full BCa bootstrap ($B = 10{,}000$) CI results for the entire 6×2×5×8 grid, stored in the `methods.ts.meta` field of each `transfer_result.json`.

References

[1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," *NeurIPS*, 2019. arXiv:1906.02530.

[2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, "On calibration of modern neural networks," *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.

[3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., "PTB-XL, a large publicly available electrocardiography dataset," *Scientific Data*, 7:154, 2020. doi:10.1038/s41597-020-0495-6.

[4] J. Zheng, J. Guo, H. Chu, "A large scale 12-lead electrocardiogram database for arrhythmia study (Chapman-Shaoxing e-cardiology)," version 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

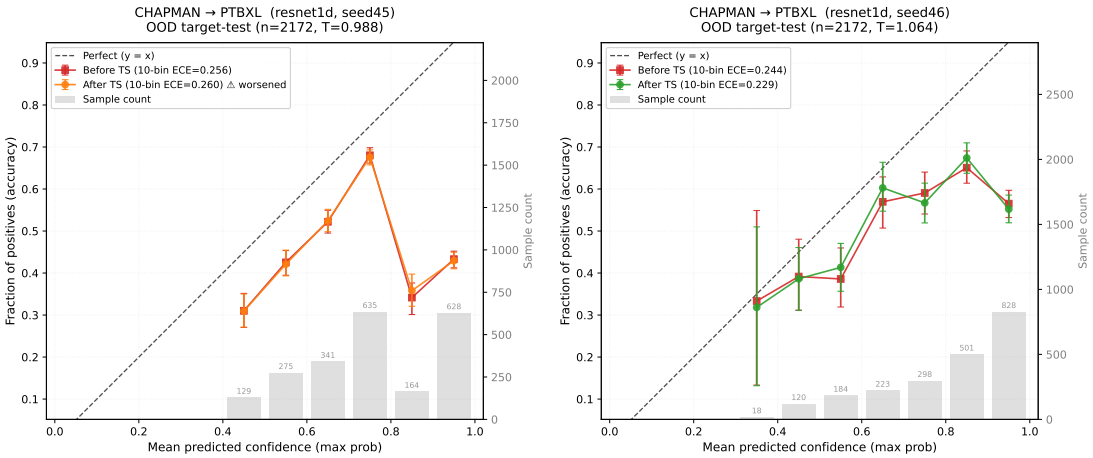


Figure 6: Chapman→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

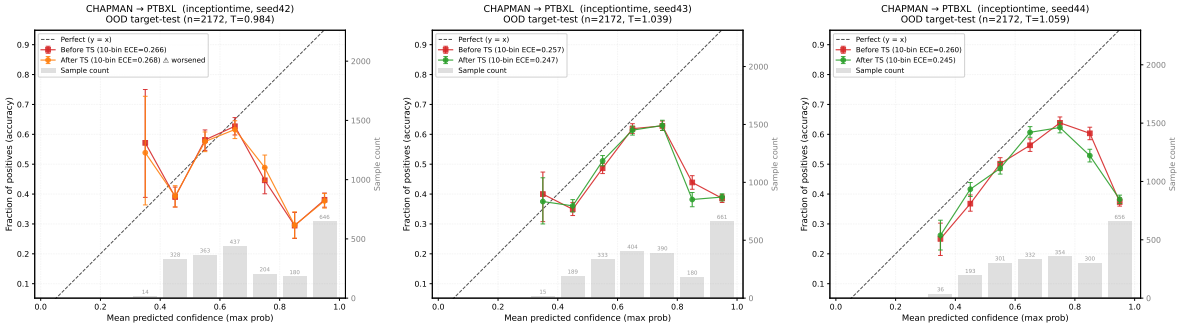


Figure 7: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

[5] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals," *Circulation*, 101(23):e215–e220, 2000.

[6] J. Nixon, M. W. Dusenberry, L. Zhang, et al., "Measuring calibration in deep learning," *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.

[7] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach, "Beyond temperature scaling: obtaining well-calibrated multi-class probabilities with Dirichlet calibration," *NeurIPS*, 2019. arXiv:1910.12656.

[8] A. Alexandari, A. Kundaje, A. Shrikumar, "Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation," *ICML*, PMLR 119:222–232, 2020. arXiv:1901.06852.

[9] Z. Lipton, Y.-X. Wang, A. Smola, "Detecting and correcting for label shift with black box predictors," *ICML*, PMLR 80:1944–1953, 2018. arXiv:1802.03916.

[10] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão, F. Cabitza, H. Gamboa, "Evaluation of uncertainty quantification methods in multi-label classification: a case study with automatic diagnosis of electrocardiogram," *Information Fusion*, 101:101978, 2024. doi:10.1016/j.inffus.2023.101978.

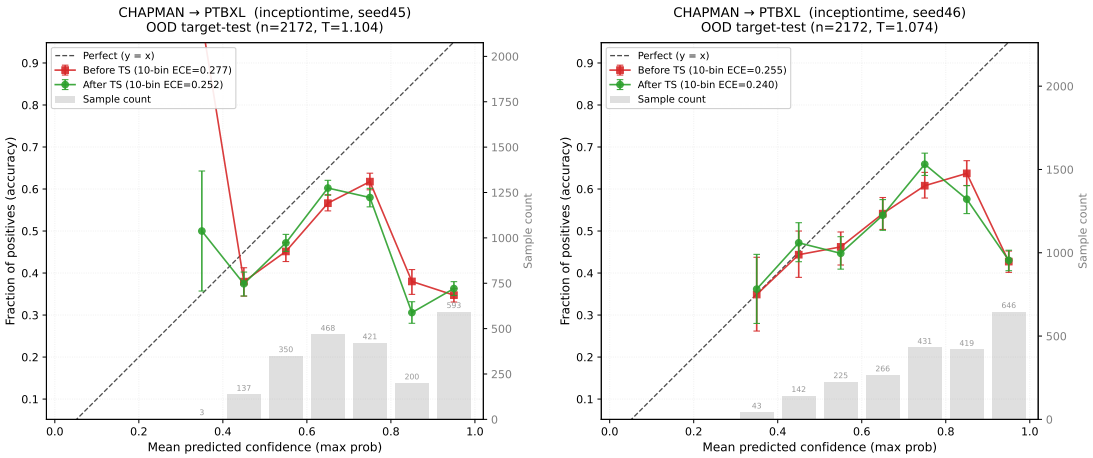


Figure 8: Chapman→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

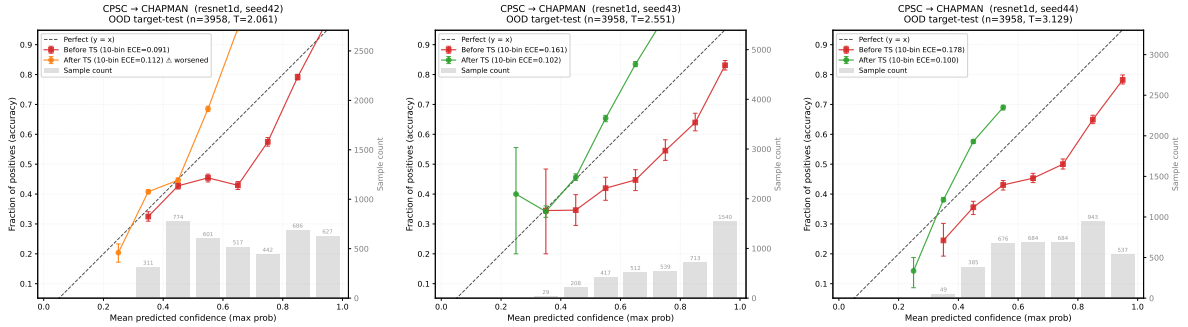


Figure 9: CPSC→Chapman reliability diagrams: ResNet1D seeds 42–44.

[11] J. F. Vranken et al., "Uncertainty estimation for deep learning-based automated analysis of 12-lead electrocardiograms," *European Heart Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.

[12] M. Reyna, N. Strodthoff, P. Wagner, et al., "Issues in the automated classification of multilead ECGs using heterogeneous labels and populations," *Physiological Measurement*, 43(8):084001, 2022. doi:10.1088/1361-6579/ac79fd.

[13] B. Efron, "Better bootstrap confidence intervals," *Journal of the American Statistical Association*, 82(397):171–185, 1987. doi:10.1080/01621459.1987.10478410.

[14] B. Van Calster, D. Nieboer, Y. Vergouwe, B. De Cock, M. J. Pencina, E. W. Steyerberg, "A calibration hierarchy for risk models was defined: from utopia to empirical data," *Journal of Clinical Epidemiology*, 74:167–176, 2016. doi:10.1016/j.jclinepi.2015.12.005.

[15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, 17:230, 2019. doi:10.1186/s12916-019-1466-7.

[16] A. Kumar, P. Liang, T. Ma, "Verified uncertainty calibration," *NeurIPS*, 2019. arXiv:1909.10155.

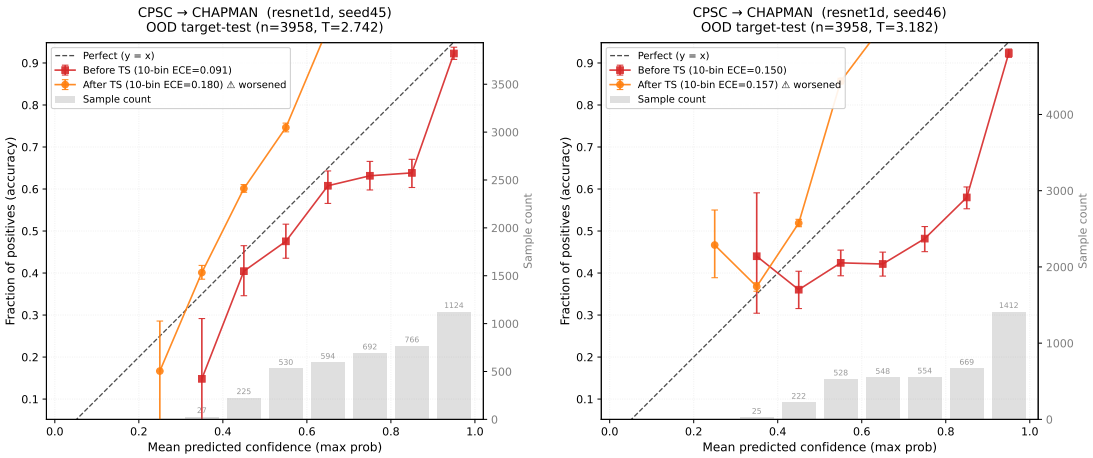


Figure 10: CPSC→Chapman reliability diagrams: ResNet1D seeds 45–46.

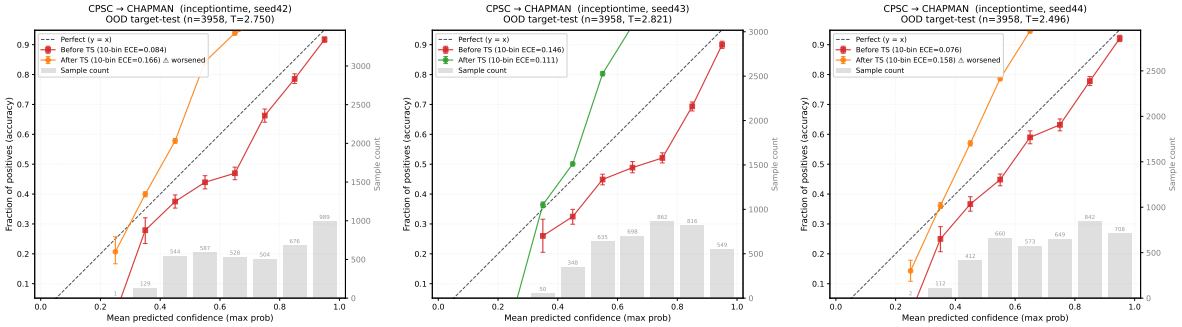


Figure 11: CPSC→Chapman reliability diagrams: InceptionTime seeds 42–44.

[17] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, "Deep learning for ECG analysis: benchmarks and insights from PTB-XL," *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. doi:10.1109/JBHI.2020.3022989.

[18] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto, "RR-interval-based atrial fibrillation detection and burden estimation: cross-dataset validation and calibration-aware probability analysis," *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.

[19] K. Patel, P. Beedala, "Calibration drift under cross-institutional deployment: an external validation framework for ICU mortality prediction across MIMIC-IV and eICU," *medRxiv* (preprint), 2026. doi:10.64898/2026.05.03.26352335.

[20] X. Wang, M. Long, J. Wang, M. I. Jordan, "TransCal: transferable calibration with lower bias and variance in domain adaptation," *NeurIPS*, 2020. arXiv:2007.08259.

[21] D. Hu, J. Liang, X. Wang, C.-S. Foo, "Pseudo-Calibration: improving predictive uncertainty estimation in unsupervised domain adaptation," *ICML*, 2024, PMLR 235:19304–19326.

[22] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, "A unified view of label shift estimation," *NeurIPS*, 2020. arXiv:2003.07554.

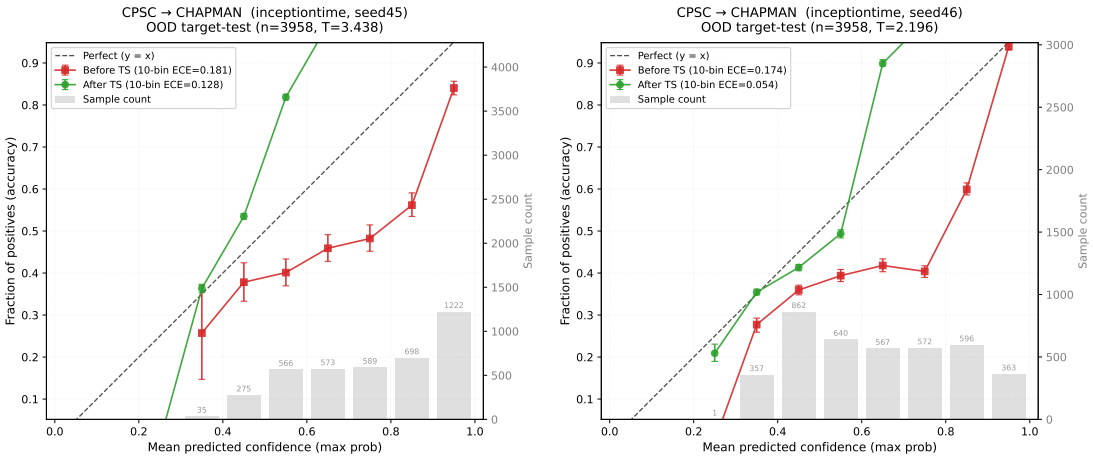


Figure 12: CPSC→Chapman reliability diagrams: InceptionTime seeds 45–46.

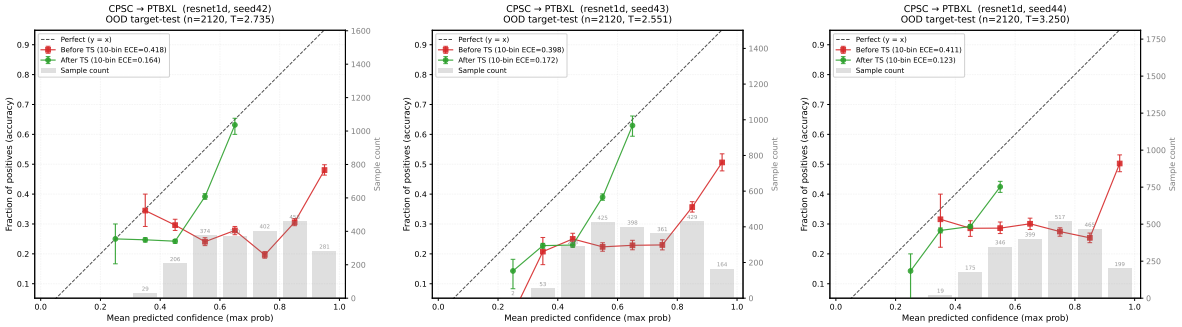


Figure 13: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 42–44.

[23] H. Zhang, H. Singh, M. Ghassemi, S. Joshi, "Why did the model fail?" Attributing model performance changes to distribution shifts," *ICML*, 2023, PMLR 202:41550–41578.

[24] T. Popordanoska, G. Radevski, T. Tuytelaars, M. B. Blaschko, "LaSCal: Label-Shift Calibration without target labels," *NeurIPS*, 2024.

[25] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.

[26] [Pre-registered protocol, publicly archived] "ECG calibration boundary study – experiment protocol," 2026. Immutable public archive with per-file SHA-256 checksums: `https://doi.org/10.17605/OSF.IO/53H64` (including revision A1 full text).

[27] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, "The preregistration revolution," *PNAS*, 115(11):2600–2606, 2018. doi:10.1073/pnas.1708274114.

[28] D. Lakens, "The practical alternative to the p value is the correctly used p value," *Perspectives on Psychological Science*, 16(3):639–648, 2021. doi:10.1177/1745691620958012.

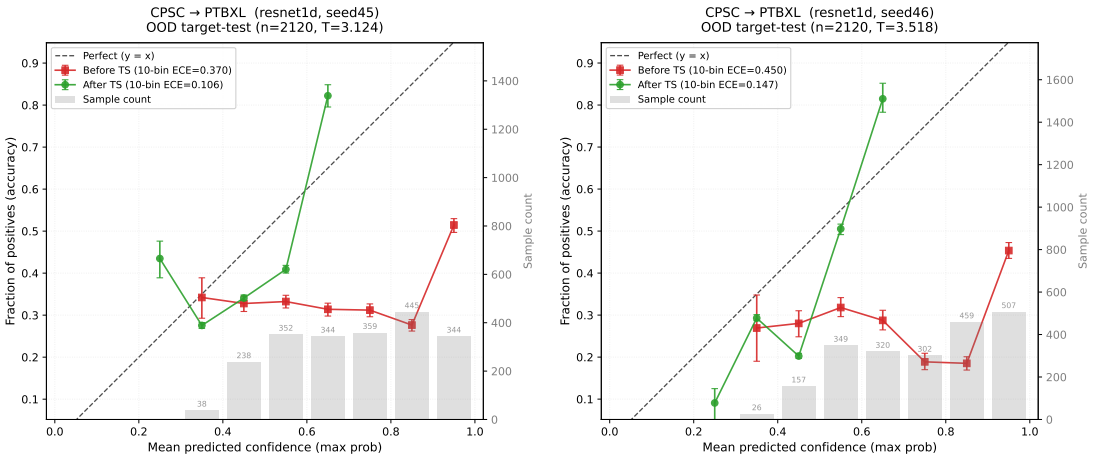


Figure 14: CPSC→PTB-XL reliability diagrams: ResNet1D seeds 45–46.

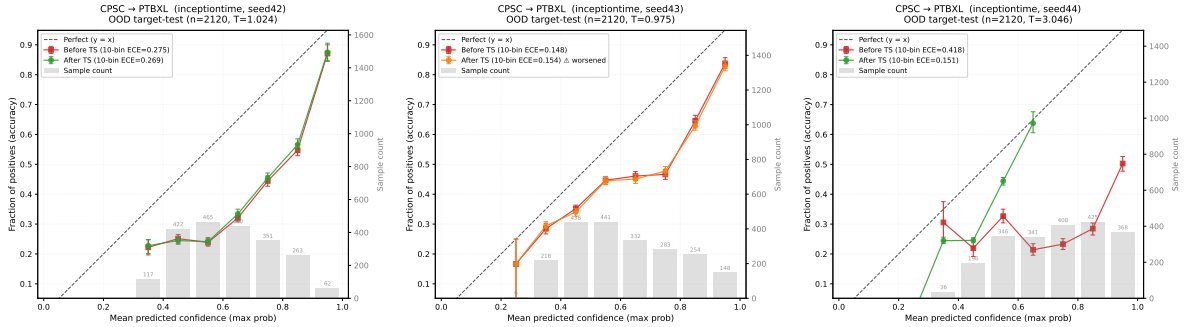


Figure 15: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 42–44.

[29] J. Li, A. D. Aguirre, V. M. Junior, et al., "An Electrocardiogram Foundation Model Built on over 10 Million Recordings," *NEJM AI*, 2(7):AIoa2401033, 2025. doi:10.1056/AIoa2401033.

[30] J. C. McHugh, B. Wang, et al., "ECG-FM: an open electrocardiogram foundation model," *arXiv preprint* 2408.05178, 2024.

[31] J. Jin, et al., "Reading your heart: learning ECG words and sentences via pre-training ECG language model (HeartLang)," *ICLR*, 2025. arXiv:2502.10707.

[32] C. V. Nguyen, C. D. Do, "Transfer learning in ECG diagnosis: is it effective?" *PLOS ONE*, 20(5):e0316043, 2025. doi:10.1371/journal.pone.0316043.

[33] S. Vavaroutas, et al., "ADAPTOOD: uncertainty-aware fine-tuning for out-of-distribution ECG time series models," *arXiv preprint* 2606.04164, 2026.

[34] W. Jiang, et al., "Test-time adaptation for ECG classification via SQI-gated self-training and beat-rhythm consistency," *arXiv preprint* 2608.23347, 2026 (accepted at MICCAI 2026).

[35] N. Nemati, "Comparative analysis of data augmentation for clinical ECG classification with STAR (Sinusoidal Time–Amplitude Resampling)," *arXiv preprint* 2510.24740, 2025.

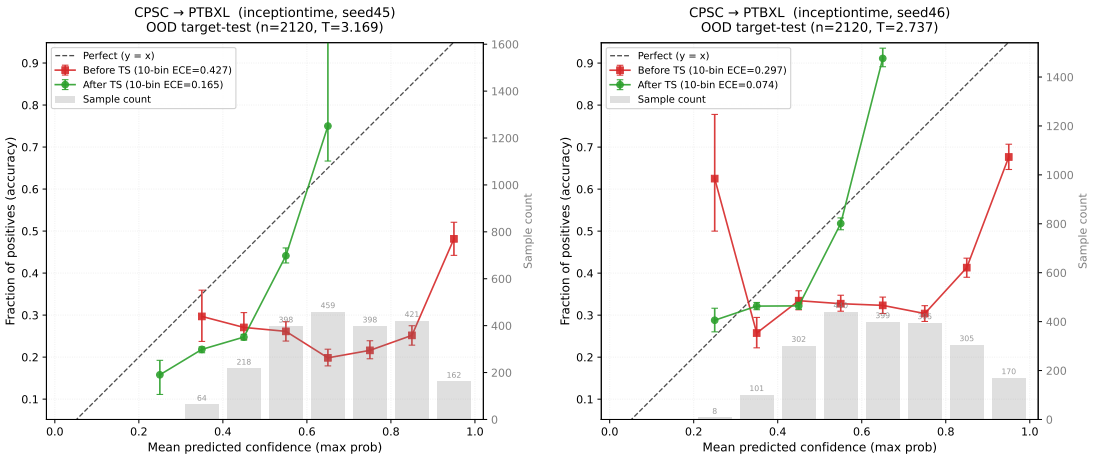


Figure 16: CPSC→PTB-XL reliability diagrams: InceptionTime seeds 45–46.

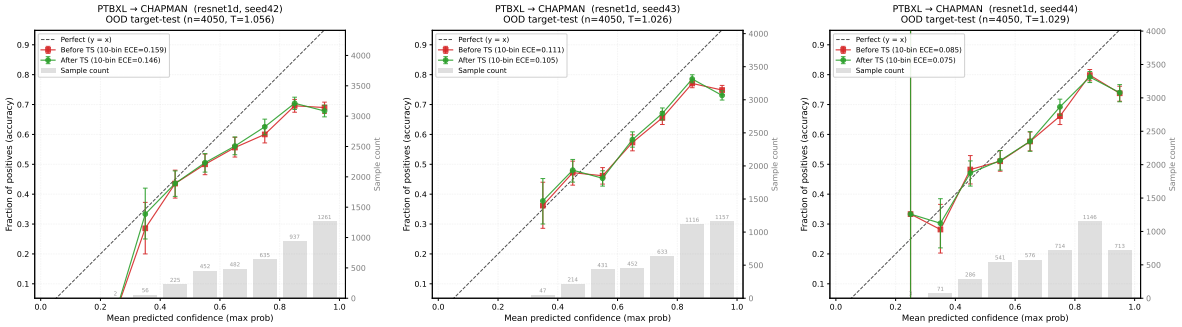


Figure 17: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 42–44.

[36] X. Liu, et al., "PEACE: knowledge-guided cross-modal fusion for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment," *arXiv preprint* 2607.15928, 2026.

[37] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, "Foundation models for electrocardiograms," *arXiv preprint* 2407.07110, 2024.

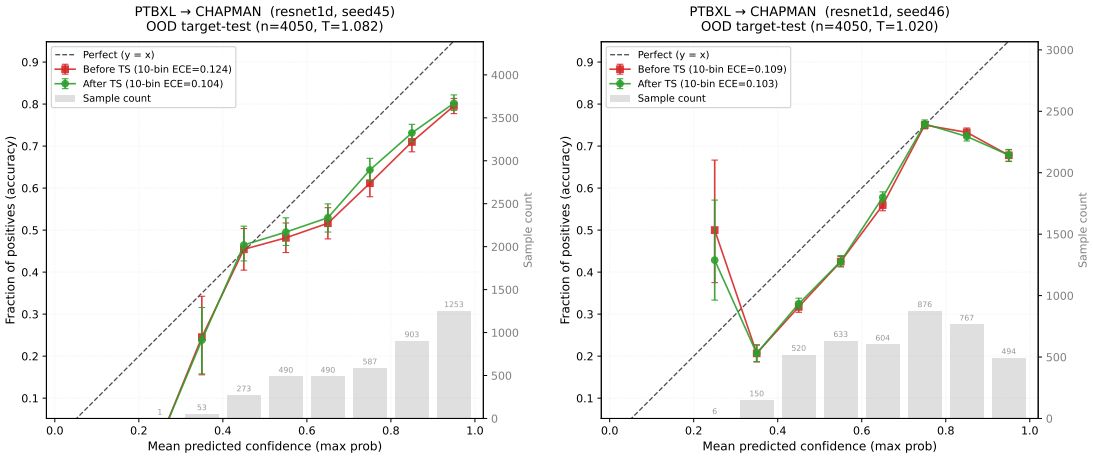


Figure 18: PTB-XL→Chapman reliability diagrams: ResNet1D seeds 45–46.

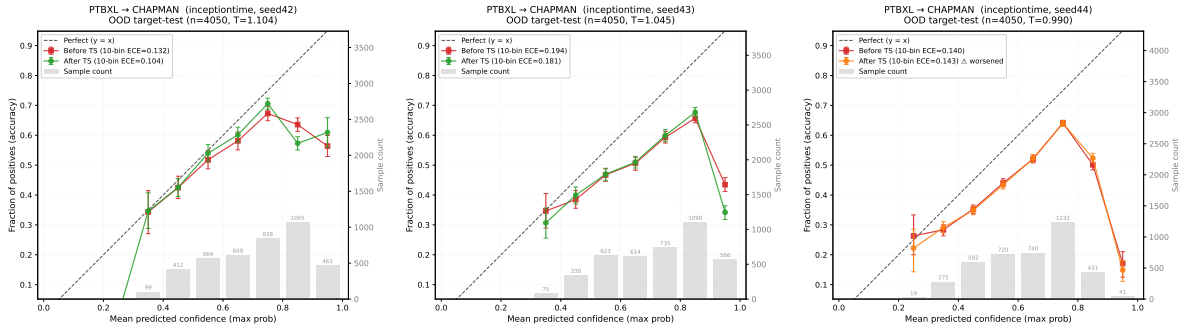


Figure 19: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 42–44.

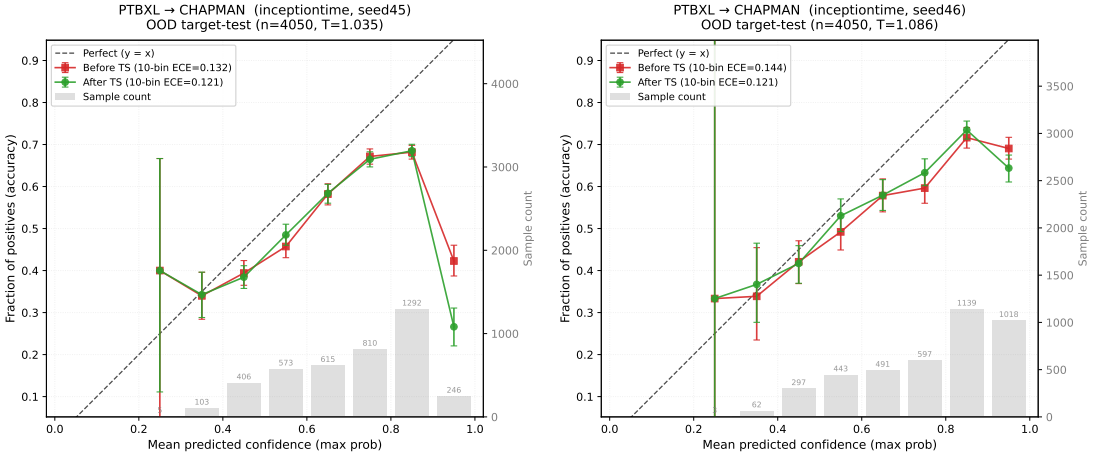


Figure 20: PTB-XL→Chapman reliability diagrams: InceptionTime seeds 45–46.

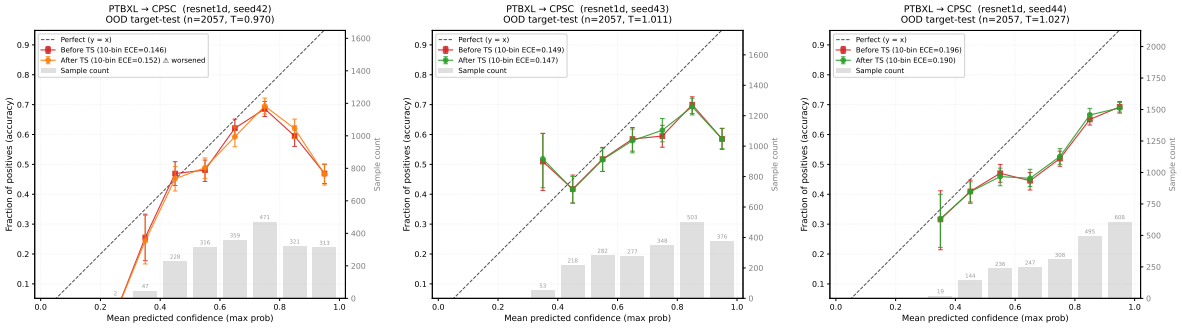


Figure 21: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 42–44.

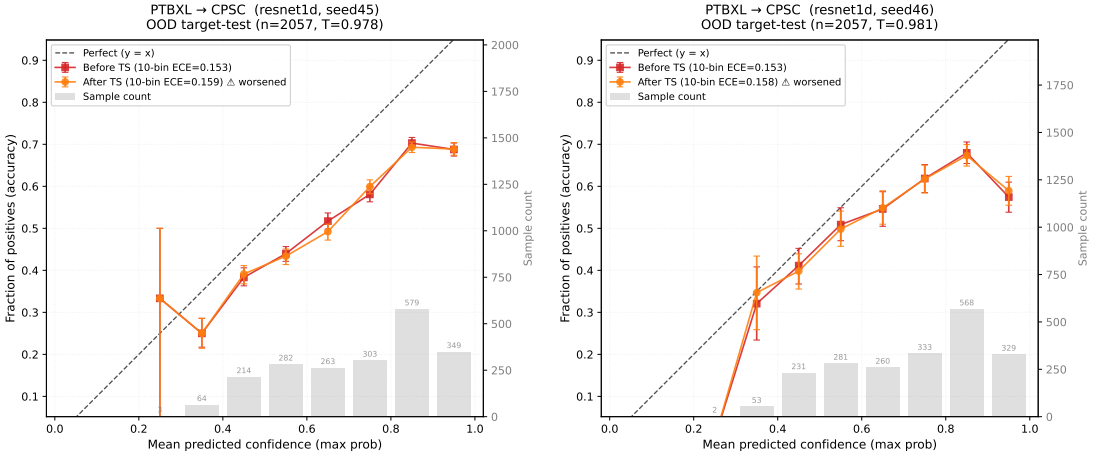


Figure 22: PTB-XL→CPSC reliability diagrams: ResNet1D seeds 45–46.

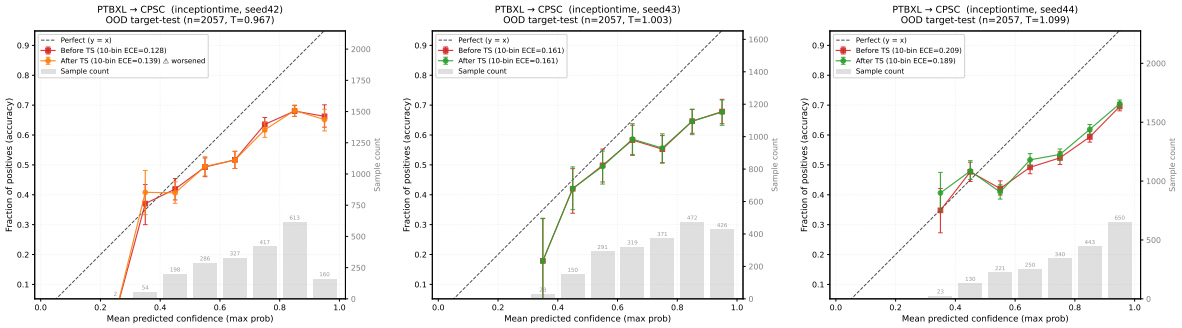


Figure 23: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 42–44.

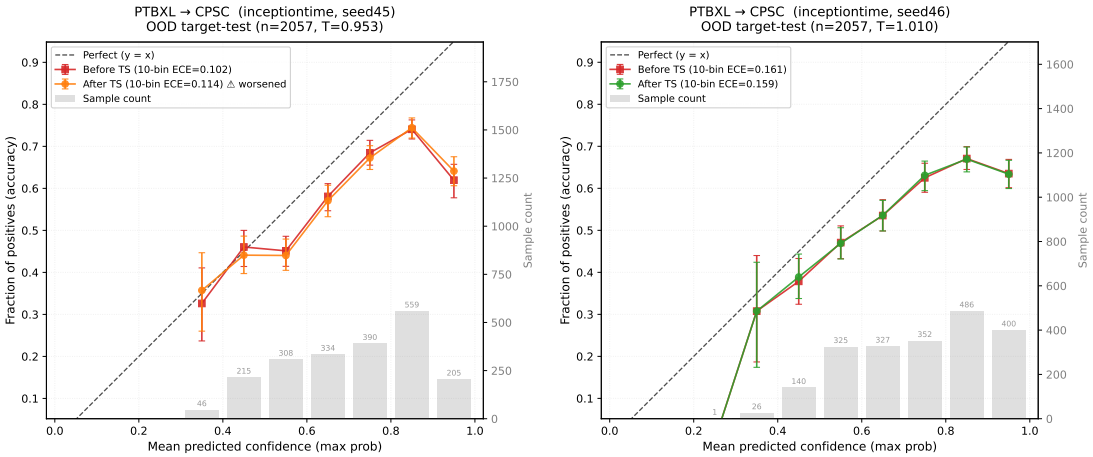


Figure 24: PTB-XL→CPSC reliability diagrams: InceptionTime seeds 45–46.

S2. Per-direction reliability diagrams and per-seed endpoint table

The three figure panels below are the main-text per-direction reliability diagrams (OOD target-test set, ResNet1D, seed 42); the full 60 per-seed catalog is Section S1. Table S1 then lists the per-seed values of the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ underlying the main-text stratum summary (Table 2 of the main text): 60 rows, columns as in the main-text table; "Support" = 95% CI lower bound > 0.

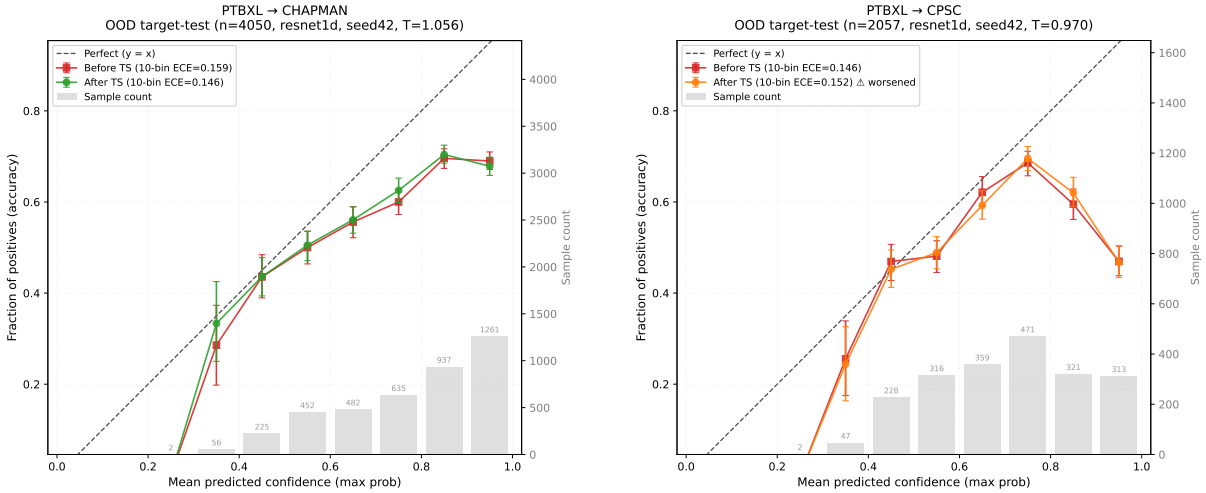


Figure 25: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

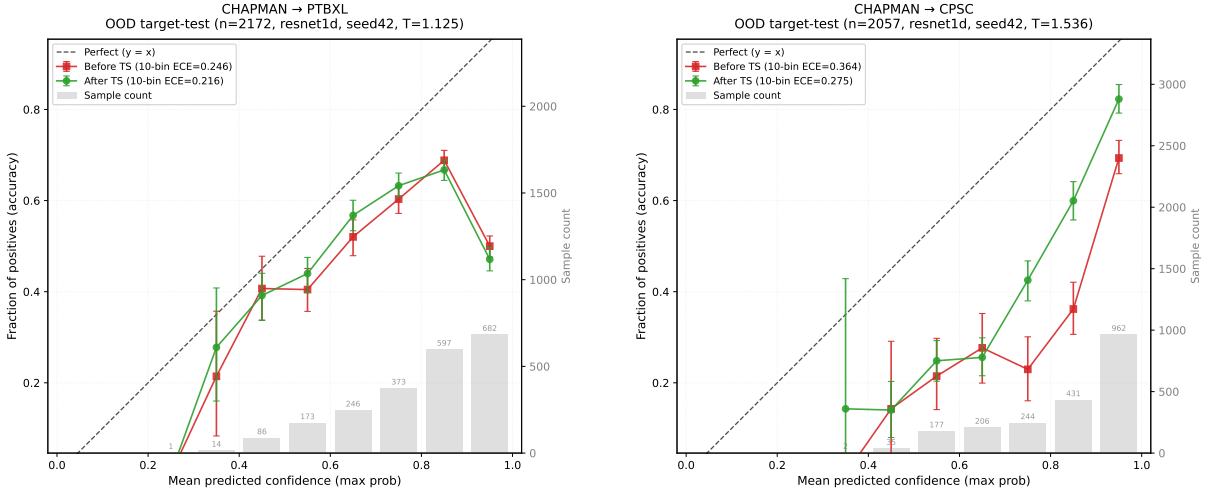


Figure 26: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

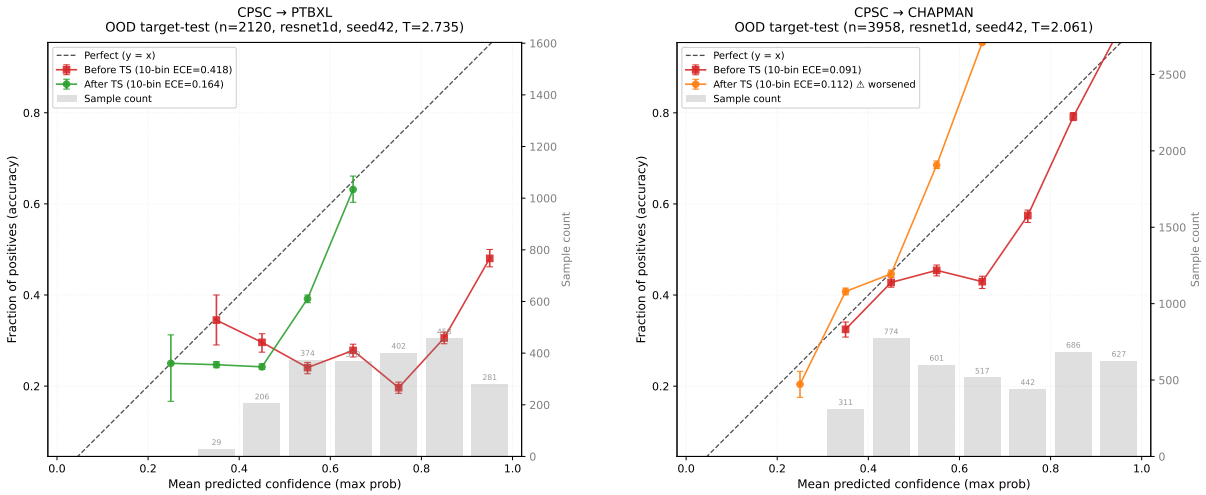


Figure 27: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

Table 1: Supplementary Table S1. Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs × 2 architectures × 5 seeds, B = 10,000, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC and $K = 5$ for the two CPSC-free pairs; see the main-text shift-matrix section. "Support" = 95% CI lower bound > 0. Fifteen counter-examples are transparently disclosed. Twenty rows (the two PTB-XL-source pairs, one per architecture per seed) were retrained under the repaired split semantics; the other 40 rows are untouched. Full 5-seed coverage on both architectures for all 6 directions.

Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC (K=4)					
IT	42	+0.0075	+0.0139	+0.0135	✓
IT	43	+0.0012	+0.0101	+0.0099	✓
IT	44	+0.0104	+0.0186	+0.0181	✓
IT	45	+0.0019	−0.0056	−0.0058	✗
IT	46	+0.0044	+0.0130	+0.0126	✓
RN	42	+0.0002	+0.0205	+0.0200	✓
RN	43	+0.0003	−0.0006	−0.0006	✗
RN	44	−0.0007	+0.0078	+0.0075	✓
RN	45	−0.0016	+0.0271	+0.0263	✓
RN	46	−0.0024	+0.0169	+0.0163	✓
Chap→PTB-XL (K=5)					
IT	42	+0.0004	−0.0025	−0.0035	✗
IT	43	+0.0005	+0.0095	+0.0068	✓
IT	44	+0.0055	+0.0136	+0.0106	✓
IT	45	−0.0128	+0.0238	+0.0208	✓
IT	46	+0.0048	+0.0162	+0.0141	✓
RN	42	−0.0031	+0.0309	+0.0291	✓
RN	43	+0.0128	+0.0291	+0.0284	✓
RN	44	+0.0059	+0.0172	+0.0168	✓
RN	45	−0.0010	−0.0036	−0.0036	✗
RN	46	−0.0042	+0.0149	+0.0140	✓
CPSC→Chap (K=4)					

Table 1 (continued)

Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
IT	42	+0.0029	+0.0065	+0.0061	✓
IT	43	−0.0012	+0.0209	+0.0203	✓
IT	44	+0.0164	+0.0182	+0.0167	✓
IT	45	+0.0091	+0.0245	+0.0242	✓
IT	46	+0.0014	+0.0024	+0.0024	✓
RN	42	−0.0002	−0.0010	−0.0012	✗
RN	43	+0.0038	+0.0096	+0.0094	✓
RN	44	−0.0152	−0.0168	−0.0170	✗
RN	45	+0.0137	+0.0181	+0.0145	✓
RN	46	+0.0017	+0.0017	+0.0016	✓
CPSC→PTB-XL (K=4)					
IT	42	+0.0028	+0.0058	+0.0056	✓
IT	43	−0.0038	−0.0057	−0.0058	✗
IT	44	+0.0158	+0.0195	+0.0190	✓
IT	45	+0.0052	+0.0197	+0.0193	✓
IT	46	+0.0002	+0.0002	+0.0002	✓
RN	42	+0.0114	+0.0150	+0.0148	✓
RN	43	+0.0048	−0.0050	−0.0051	✗
RN	44	−0.0032	−0.0049	−0.0049	✗
RN	45	+0.0130	+0.0176	+0.0173	✓
RN	46	+0.0261	+0.0302	+0.0295	✓
PTB-XL→Chap (K=5)					
IT	42	+0.0183	+0.0254	+0.0212	✓
IT	43	+0.0046	+0.0125	+0.0119	✓
IT	44	−0.0025	−0.0033	−0.0034	✗
IT	45	+0.0062	+0.0108	+0.0099	✓
IT	46	+0.0162	+0.0217	+0.0190	✓
RN	42	+0.0036	+0.0128	+0.0118	✓
RN	43	+0.0020	+0.0065	+0.0060	✓
RN	44	−0.0008	+0.0070	+0.0062	✓
RN	45	+0.0125	+0.0190	+0.0175	✓
RN	46	+0.0040	+0.0058	+0.0057	✓
PTB-XL→CPSC (K=4)					
IT	42	+0.0052	−0.0083	−0.0089	✗
IT	43	+0.0002	+0.0006	+0.0005	✓
IT	44	+0.0161	+0.0209	+0.0198	✓
IT	45	−0.0056	−0.0123	−0.0132	✗
IT	46	+0.0002	+0.0024	+0.0022	✓
RN	42	−0.0057	−0.0069	−0.0079	✗
RN	43	+0.0018	+0.0024	+0.0021	✓
RN	44	+0.0019	+0.0063	+0.0060	✓
RN	45	−0.0036	−0.0057	−0.0059	✗
RN	46	−0.0022	−0.0048	−0.0051	✗